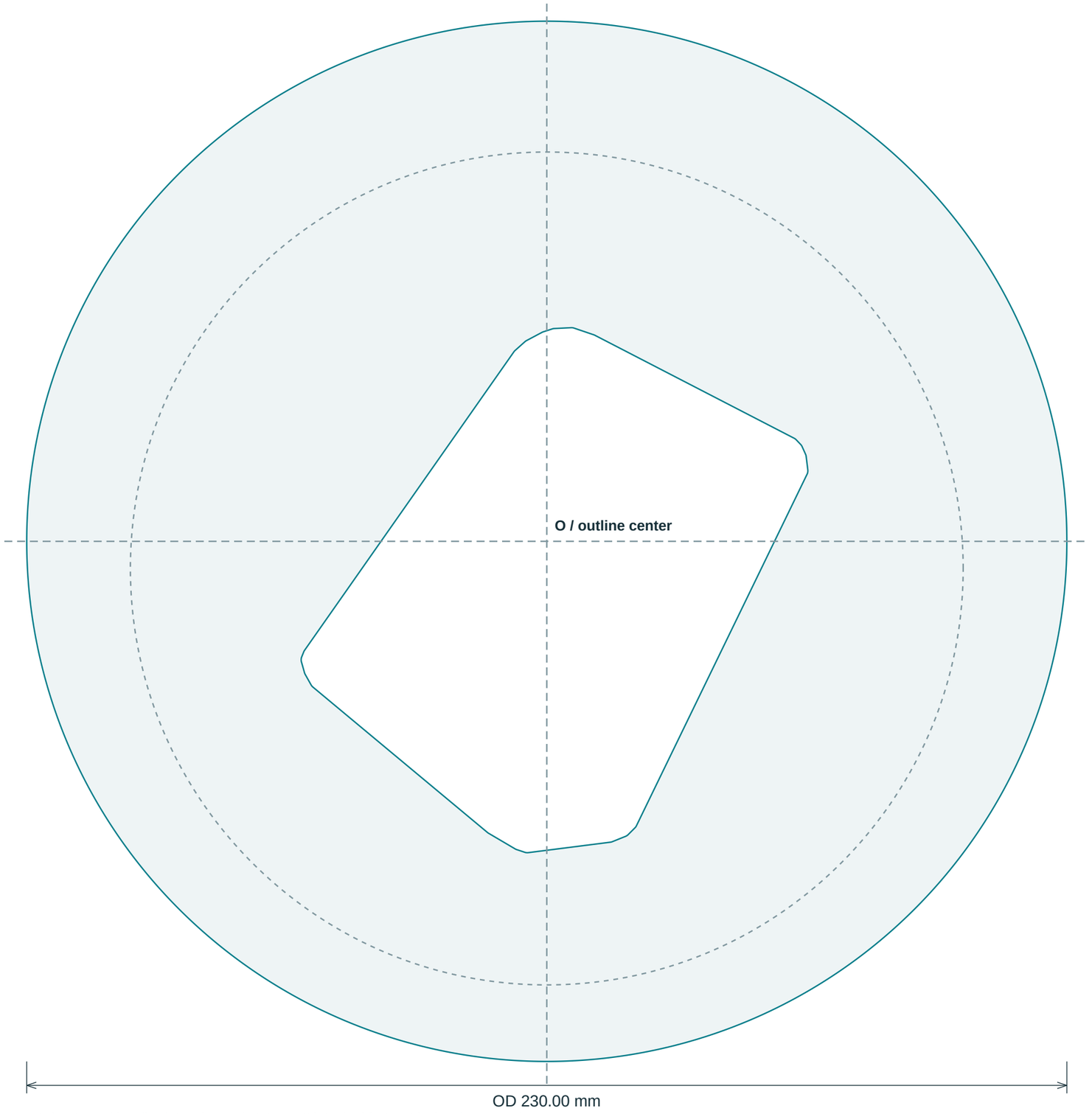
MAIN-A / Offset round main plate

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION



Front view toward handles | Drawing scale 1:1 on A3 | Dashed factory trim is reference only

PART DEFINITION

6 mm upward offset from inferred main trim center creates a gap to the matching diverter plate.

Item	Specification
Origin O	Center of custom outer outline; rear face Z = 0.
Installed O	X 0.799; Y 45.673 mm (conditional scan frame).
Outer profile	Circle OD 230.00
Aperture	AP-MAIN: see sheet 5 / coordinate file
Separate holes	None. No countersink, thread or added bolt hole.
Cutting file	MAIN-A_F005_CUT_ONLY_DRAFT_NOT_FOR_FABRICATION.dxf
Plan area	337.1 cm2
Total cut length	1062.4 mm

CHOOSE ONE STOCK OPTION

Suffix	Material / exact CAD thickness
SS060	316 stainless / 1.524 mm (0.060 inch)
SS125	316 stainless / 3.175 mm (0.125 inch)
PROTO3	Dry-fit plastic / 3.000 mm

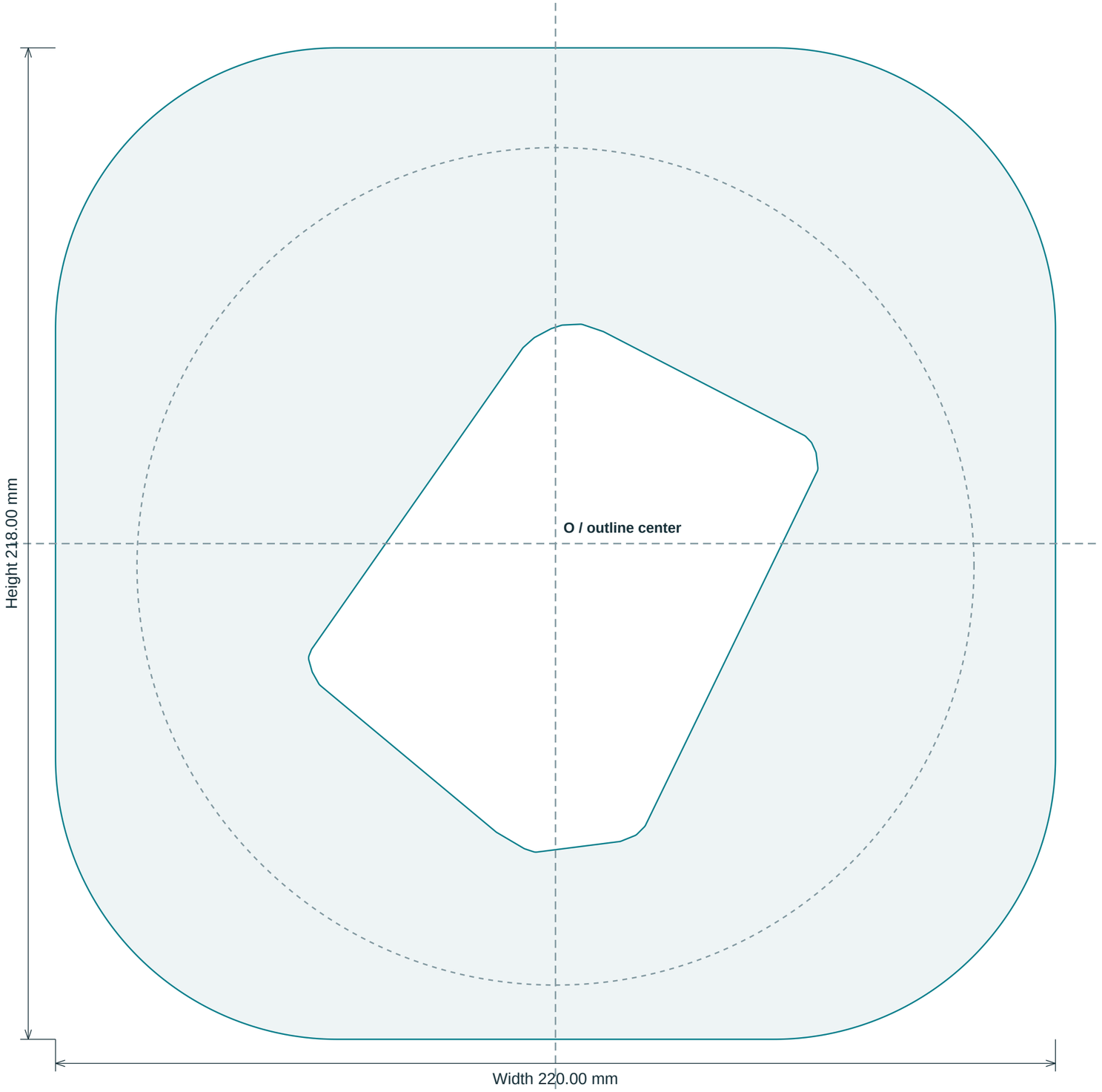
Thickness is a separate order input. The DXF profile is common to all three thicknesses. Select one suffix; do not order every file.

Quote target: XY +/-0.13 mm; global flatness <=0.5 mm, subject to shop acceptance. Actual stock thickness range must be supplied. See sheet 8 for finish, support and release conditions.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

MAIN-B / Rounded rectangular main plate

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION



Front view toward handles | Drawing scale 1:1 on A3 | Dashed factory trim is reference only

PART DEFINITION

Alternative softer rectangular outline; 5 mm upward offset from inferred trim center.

Item	Specification
Origin O	Center of custom outer outline; rear face Z = 0.
Installed O	X 0.799; Y 44.673 mm (conditional scan frame).
Outer profile	220.00 x 218.00; four corners R62.00
Aperture	AP-MAIN: see sheet 5 / coordinate file
Separate holes	None. No countersink, thread or added bolt hole.
Cutting file	MAIN-B_F005_CUT_ONLY_DRAFT_NOT_FOR_FABRICATION.dxf
Plan area	368.2 cm2
Total cut length	1109.4 mm

CHOOSE ONE STOCK OPTION

Suffix	Material / exact CAD thickness
SS060	316 stainless / 1.524 mm (0.060 inch)
SS125	316 stainless / 3.175 mm (0.125 inch)
PROTO3	Dry-fit plastic / 3.000 mm

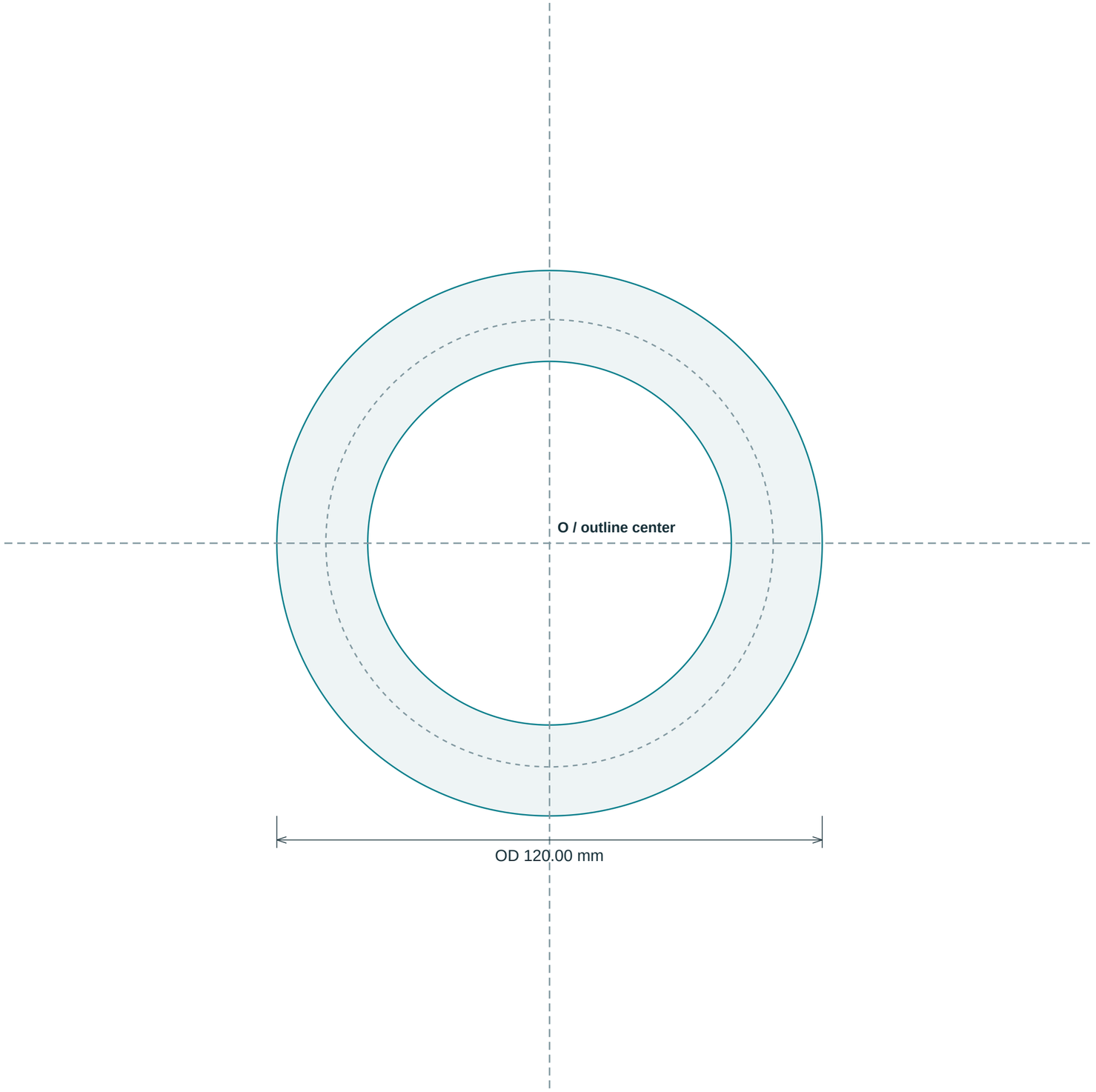
Thickness is a separate order input. The DXF profile is common to all three thicknesses. Select one suffix; do not order every file.

Quote target: XY +/-0.13 mm; global flatness <=0.5 mm, subject to shop acceptance. Actual stock thickness range must be supplied. See sheet 8 for finish, support and release conditions.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

DIV-A / Round diverter companion

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION



Front view toward handles | Drawing scale 1:1 on A3 | Dashed factory trim is reference only

PART DEFINITION

Matching cosmetic/support ring. Factory diverter plate assembly and rear sealing footprint remain unmeasured; diameter 80 is a clearance trial.

Item	Specification
Origin O	Center of custom outer outline; rear face Z = 0.
Installed O	X -0.067; Y -135.563 mm (conditional scan frame).
Outer profile	Circle OD 120.00
Aperture	Concentric through opening OD 80.00 mm, trial dimension.
Separate holes	None. No countersink, thread or added bolt hole.
Cutting file	DIV-A_F005_CUT_ONLY_DRAFT_NOT_FOR_FABRICATION.dxf
Plan area	62.8 cm2
Total cut length	628.3 mm

CHOOSE ONE STOCK OPTION

Suffix	Material / exact CAD thickness
SS060	316 stainless / 1.524 mm (0.060 inch)
SS125	316 stainless / 3.175 mm (0.125 inch)
PROTO3	Dry-fit plastic / 3.000 mm

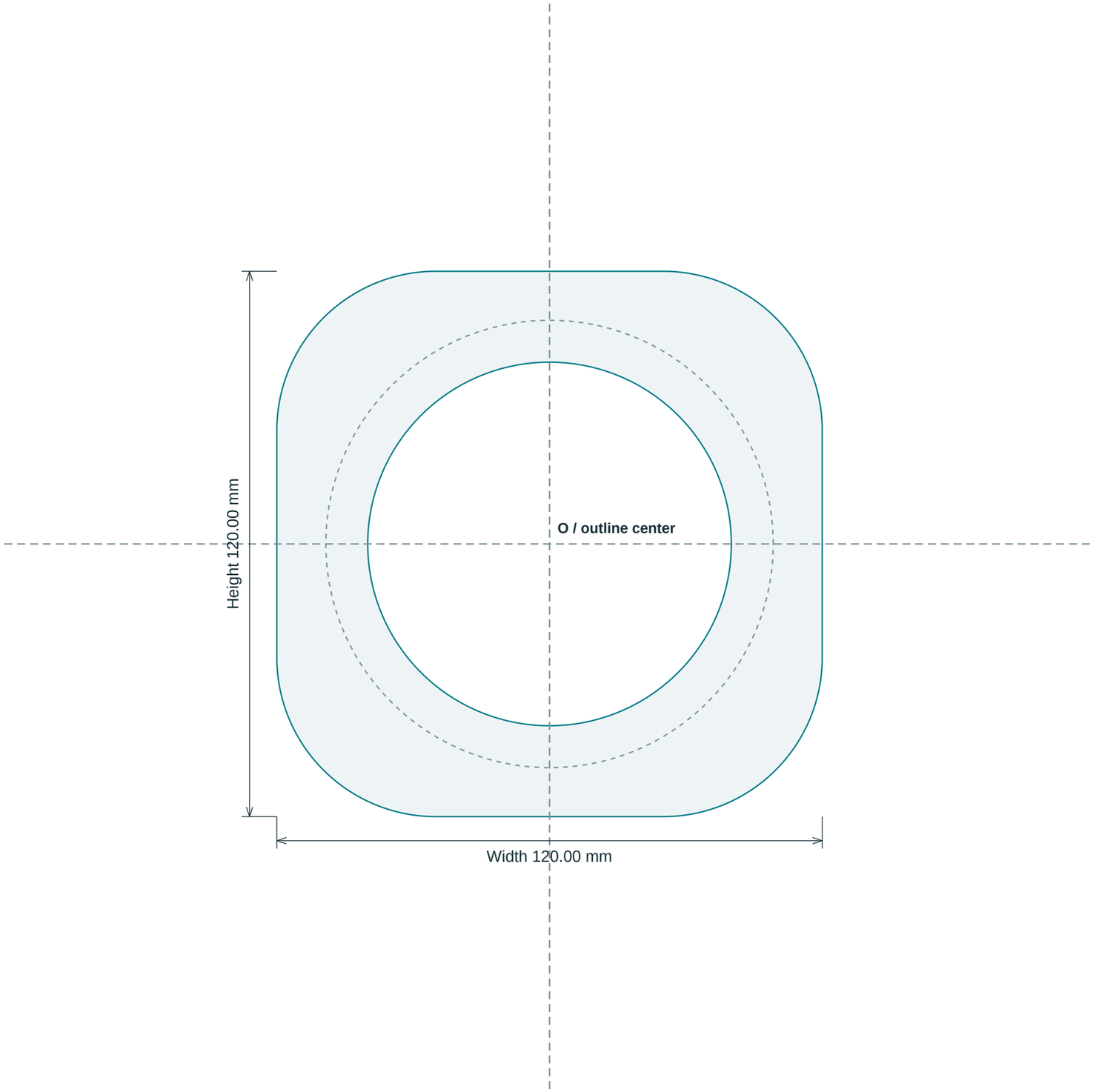
Thickness is a separate order input. The DXF profile is common to all three thicknesses. Select one suffix; do not order every file.

Quote target: XY +/-0.13 mm; global flatness <=0.5 mm, subject to shop acceptance. Actual stock thickness range must be supplied. See sheet 8 for finish, support and release conditions.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

DIV-B / Rounded square diverter companion

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION



Front view toward handles | Drawing scale 1:1 on A3 | Dashed factory trim is reference only

PART DEFINITION

Companion to MAIN-B; same conditional 80 mm central aperture.

Item	Specification
Origin O	Center of custom outer outline; rear face Z = 0.
Installed O	X -0.067; Y -135.563 mm (conditional scan frame).
Outer profile	120.00 x 120.00; four corners R35.00
Aperture	Concentric through opening OD 80.00 mm, trial dimension.
Separate holes	None. No countersink, thread or added bolt hole.
Cutting file	DIV-B_F005_CUT_ONLY_DRAFT_NOT_FOR_FABRICATION.dxf
Plan area	83.2 cm2
Total cut length	671.2 mm

CHOOSE ONE STOCK OPTION

Suffix	Material / exact CAD thickness
SS060	316 stainless / 1.524 mm (0.060 inch)
SS125	316 stainless / 3.175 mm (0.125 inch)
PROTO3	Dry-fit plastic / 3.000 mm

Thickness is a separate order input. The DXF profile is common to all three thicknesses. Select one suffix; do not order every file.

Quote target: XY +/-0.13 mm; global flatness <=0.5 mm, subject to shop acceptance. Actual stock thickness range must be supplied. See sheet 8 for finish, support and release conditions.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

One continuous aperture clears both control assemblies and the captured stud paths. No separate fastener holes are added.

The cut contour is the following closed convex polygon offset OUTWARD by 1.75 mm, with circular arc joins. The DXF and STEP contain that exact offset. Coordinates below are local to each outer-outline center O. They are design coordinates derived from the conditional scan, not accepted caliper readings. Do not scale a screenshot.

Vertex	MAIN-A X	MAIN-A Y	MAIN-B X	MAIN-B Y
1	-4.3058	-67.0944	-4.3058	-66.0944
2	-6.1100	-66.4959	-6.1100	-65.4959
3	-11.9920	-63.0345	-11.9920	-62.0345
4	-50.5918	-30.8838	-50.5918	-29.8838
5	-51.8932	-28.5696	-51.8932	-27.5696
6	-52.5645	-26.1963	-52.5645	-25.1963
7	-52.1225	-25.0966	-52.1225	-24.0966
8	-5.8368	40.8972	-5.8368	41.8972
9	-3.6629	42.8349	-3.6629	43.8349
10	-0.2500	44.6489	-0.2500	45.6489
11	1.7637	45.3037	1.7637	46.3037
12	5.4726	45.4830	5.4726	46.4830
13	9.8328	44.0019	9.8328	45.0019
14	53.8878	21.2399	53.8878	22.2399
15	54.8662	20.2143	54.8662	21.2143
16	55.6209	18.5417	55.6209	19.5417
17	55.9790	15.5954	55.9790	16.5954
18	18.2530	-62.1166	18.2530	-61.1166
19	16.7574	-63.6091	16.7574	-62.6091
20	13.8360	-64.7873	13.8360	-63.7873

DATUM AND INSTALLATION LOGIC

Rear plate face = datum A, Z=0 in each STEP. Outline center O = XY origin. Front points +Z. In the installation the plate rear is shown 1 mm above tile solely as a depth scenario. The actual cured seal/support gap must be established and the handle/adapter stack rechecked.

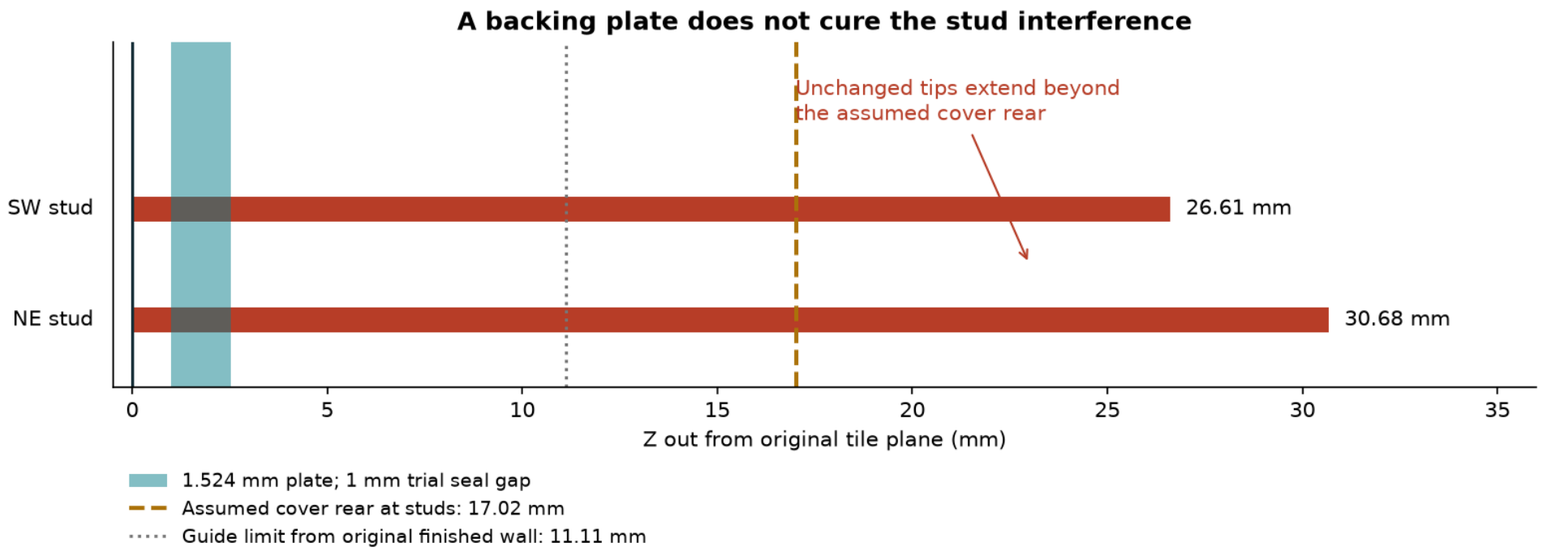
Temporary centering method: print the matching template at 100%, verify both 100 mm bars, locate the control/stud references, then mark removable tape on intact stone at the outer top and sides. Place the clear prototype to those marks. Do not use the oversized central aperture as a locating feature.

Main A is 6 mm higher than the inferred cover center; main B is 5 mm higher. This is deliberate and makes space for the diverter companion. Measured outer-cover registration could revise these offsets.

Removal path: remove factory handles/cover and inner plate as the applicable Kohler guide permits; withdraw backing plate axially. Factory studs, adapters and valve stay unchanged unless separately corrected by the installer. Confirm actual tip/head clearance before any forcing.

Captured forward surface check: 51,929 clipped triangles; 0 outside the aperture. The mask excludes a 3 mm boundary strip of stone. Unseen geometry and physical scale remain unresolved.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.



ASSEMBLY ORDER

1. Existing sound travertine and qualified support/landing repair where required.

2. Qualified wall-to-custom-plate surface seal; joint section and adhesion to be selected.

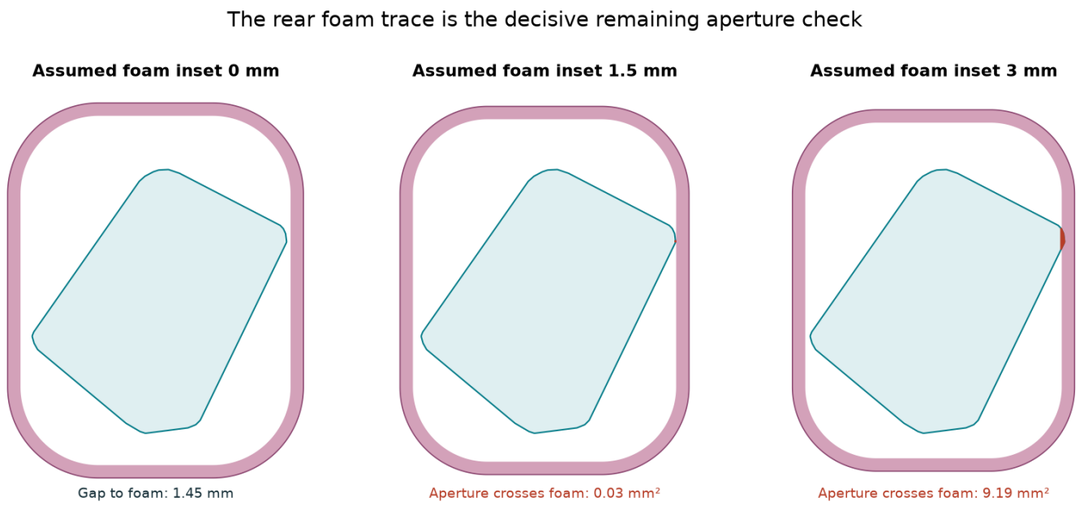
3. Custom backing plate, selected thickness.

4. Retained factory foam and inner plate; foam must land continuously.

5. Factory nuts behind decorative cover. Stud tips must not interfere.

6. Factory cover, notch down, correct adapters/bonnets/handles.

The new surface plate does not establish concealed membrane continuity. Keep the factory bottom drain route open onto the front of the custom plate.



Z values still needed for release

Stud depths/axes, local cover cavity over both nuts, rear foam footprint and installed compression, actual rear metal lands, and handle-adapter engagement. The free 15.44 mm envelope is known; the installed one is not.

The current 14.5 mm cavity assumption minus 15.44 mm free envelope is -0.94 mm. Compression can change that comparison, but no allowed compression has been established. Do not specify a generic 5 mm sealing tolerance.

No stud cut length, wood drilling or valve relocation is specified. First identify the same fasteners from both sides and verify reversible factory adjustment and retained thread engagement.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

Exploded assembly / measured and nominal components

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION



READ FROM WALL TOWARD HANDLES

1 / Captured wall, valve and studs: unchanged evidence.

2 / New main and diverter backing plates: exact custom CAD.

3 / Main foam: about 6 x 6 mm section; rear footprint illustrated, not measured.

4 / Main inner plate: measured 130.35 x 165.1 mm envelope, upper opening and mounting edge constraints; formed/rear details provisional.

5 / Nuts (obscured in this view): order documented; local seats/height unmeasured.

6 / Covers, bonnets and handles: official Kohler exterior meshes. Manufacturer mesh is not a hidden-interface specification.

Diverter: retain its factory mounting plate/foam between custom plate and decorative cover. That hidden assembly is not dimensionally reconstructed; its fit is a release check.

Exploded offsets only

Spacing here separates parts for explanation. It does not give installed Z dimensions, engagement lengths or an approved adjustment.

Main trim guide: Kohler 1216755-2-A, steps 8-15. Older guide 1041504-2-F, page 7, explicitly gives 7/16 inch maximum stud projection from finished wall. Diverter guide 1190382-2-A, steps 8-13.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

Fabricator specifications / release and inspection

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION

SCOPE

Manufacture custom backing plates only. The measured Kohler reference parts and source meshes are not replacement-part manufacturing data. Choose A or B and one thickness. No order has been authorized.

MATERIAL AND FINISH

Visible face uniform satin / directional brush, same direction on both parts; protect face during cutting and shipping. Stainless will not exactly match Kohler Vibrant Brushed Nickel. Obtain finish sample before cosmetic release. Preferred metal comparison is 316 stainless; 304 #4 stock is an alternative requiring explicit material acceptance.

EDGES AND GEOMETRY

Deburr both faces and cutout; remove tabs and sharp edges. No resizing of critical contours without drawing revision. Supply no kerf compensation in our CAD. The shop applies its process allowance. DXF has two contours only.

THICKNESS AND FLATNESS

Nominal choices: 1.524 / 3.175 mm stainless, 3.000 mm prototype. Requested XY tolerance +/-0.13 mm and global flatness <=0.5 mm are quote targets, not a guarantee. Confirm stock tolerance and finishing loss. Do not force a warped plate flat using factory nuts.

SUPPORT AND RETENTION

Provide sound, rigid bearing under the wall-facing plate where clamping loads enter the factory foam. Check rocking and flex before sealing. A silicone bead is not a substitute for verified support. Removable centering tape/template is required; no tile holes are authorized.

RELEASE CHECKLIST - OPEN

[ ] Trace the actual rear foam on the template: no portion may bridge the custom aperture or a gap in support. Record minimum supported width.

[ ] Check plate insertion, stone high spots/grout crossing, worst traced cuts and separate-plate gap on the actual wall.

[ ] Reconcile actual diagonal stud axes and depths with the measured inner plate; verify stud/nut adjustment and cover seating.

[ ] Check both controls and the diverter with correct adapters, secured bonnets/handles, full permitted travel and removal/tool access.

[ ] Approve selected alloy, thickness, finish sample, support/seal detail and vendor tolerance/lead time.

[ ] Record explicit manufacturing release against F005 or a revised design before permanent manufacture.

INCOMING PART INSPECTION

Measure outer size, aperture bounds/coordinates, thickness and flatness; check grain direction, scratches, burrs/tabs, and undamaged edges. Dry-fit before any permanent sealing. After the selected system has fully cured, distinguish pressure/supply leak testing from directed shower-water entry testing, with an observation path and stop on leakage.

Manufacturer package contains custom CAD and these sheets only. The private scan/house imagery remains in the local review and Blender project.